

# Order Quantities When Demand is Approximately Level

EOQ, sensitivity, EPQ and quantity discounts: worked examples

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# 1. The EOQ model

## Assumptions

1 constant, deterministic demand; 2 Q need not be an integer, no min/max; 3 no quantity discounts; 4 costs constant over time; 5 items independent; 6 zero (or known) lead time; 7 no shortages; 8 whole order delivered at once; 9 very long horizon.

## Costs

$$C_r = \frac{A + Qv}{Q/D} = \frac{AD}{Q} + vD \quad C_c = \bar{I}vr = \frac{Qvr}{2}$$

$$TRC(Q) = \frac{Qvr}{2} + \frac{AD}{Q}$$

## Optimum

$$\frac{dTRC}{dQ} = \frac{vr}{2} - \frac{AD}{Q^2} = 0 \Rightarrow EOQ = \sqrt{\frac{2AD}{vr}}, \quad TRC(EOQ) = \sqrt{2ADvr}$$

## Other expressions

$$T_{EOQ} = \sqrt{\frac{288A}{Dvr}} \text{ months} \quad TR = \sqrt{\frac{2vrD}{A}}$$

## Sensitivity

$$Q' = (1 + \alpha)EOQ \Rightarrow PCP = 50 \frac{\alpha^2}{1 + \alpha}$$

## EPQ

$$TRC(Q) = \frac{AD}{Q} + \frac{Q(1 - D/m)vr}{2} \quad EPQ = \frac{EOQ}{\sqrt{1 - D/m}}$$

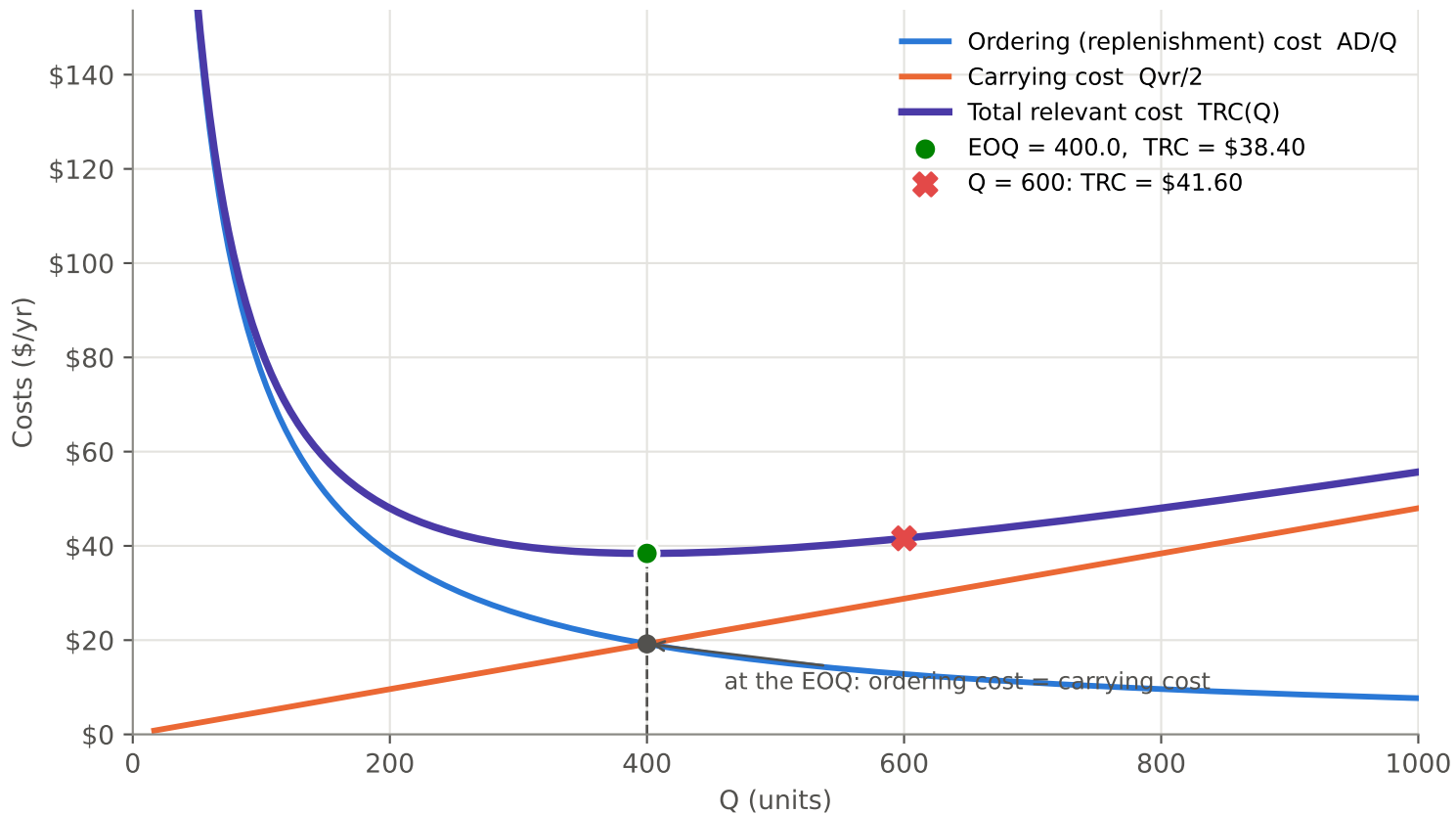
## All-units discount

Step 1: EOQ(d); if EOQ(d)  $\geq$  Q1 it is optimal (Case 3). Otherwise compare TRC(EOQ no discount) with TRC(Q1): the smaller one wins (Case 2 or Case 1).

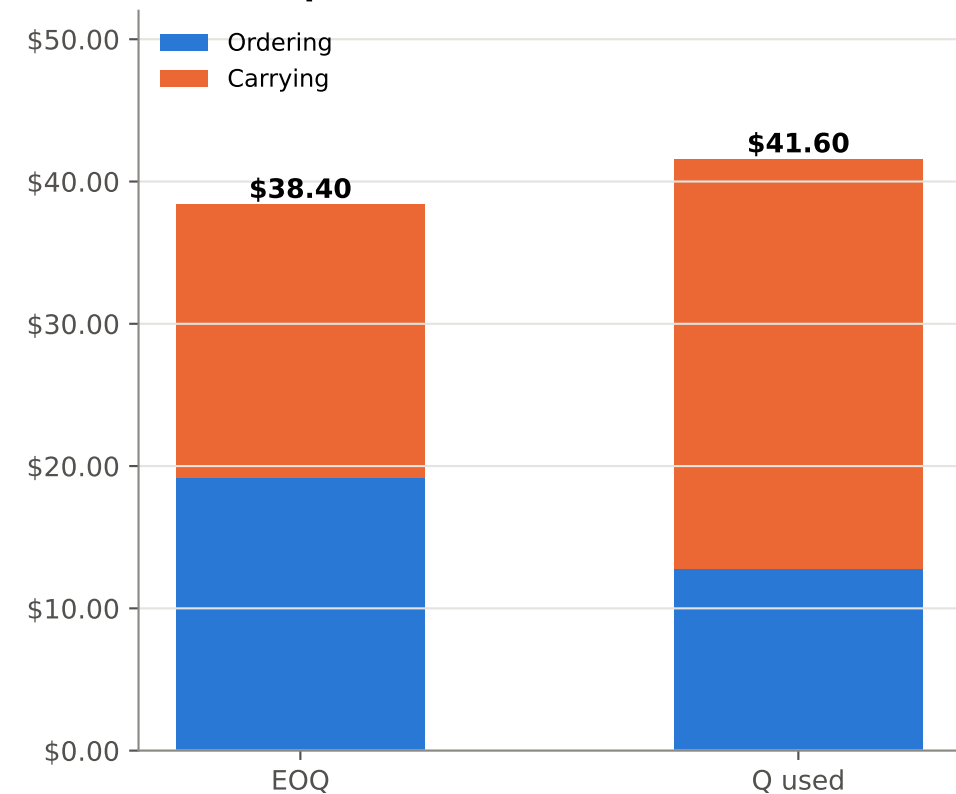
## Worked example 1: EOQ ( $A=3.2$ , $D=2400$ , $v=0.40$ , $r=0.24$ , $Q_u$ )

| Quantity                          | Value  |
|-----------------------------------|--------|
| EOQ (units)                       | 400    |
| TRC(EOQ) (\$/yr)                  | 38.4   |
| Ordering cost $AD/Q$ (\$/yr)      | 19.2   |
| Carrying cost $Qvr/2$ (\$/yr)     | 19.2   |
| Purchase cost $vD$ (\$/yr)        | 960    |
| Orders per year $D/Q$             | 6      |
| Time between orders $Q/D$ (years) | 0.1667 |
| $T_{EOQ}$ (months of supply)      | 2      |
| Time between orders (weeks)       | 8.667  |
| Average inventory $Q/2$ (units)   | 200    |
| Average inventory value (\$)      | 80     |
| Turnover ratio $TR = D/I_{bar}$   | 12     |
| $Q$ used (units)                  | 600    |
| $\alpha = Q'/EOQ - 1$             | 0.5    |
| TRC( $Q$ used) (\$/yr)            | 41.6   |
| PCP of $Q$ used (%)               | 8.333  |

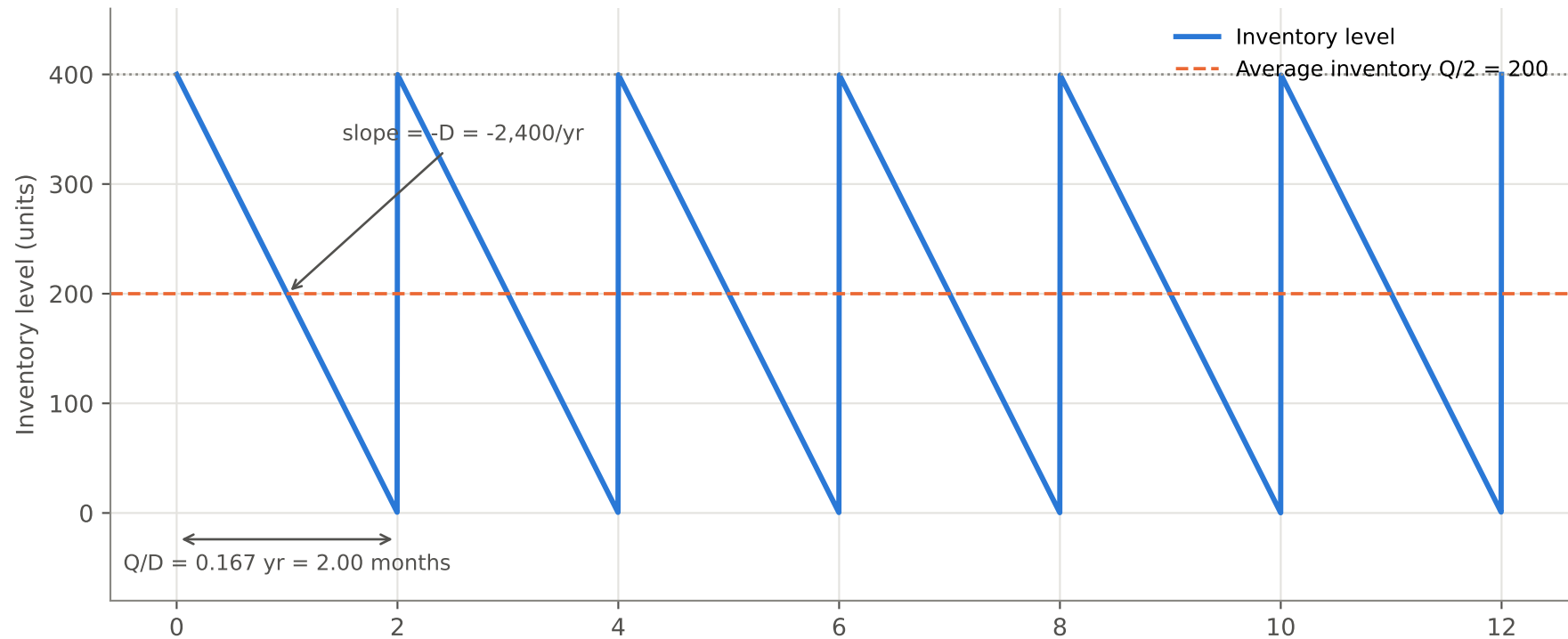
## TRC(Q): EOQ = 400.0



## Cost components (PCP of Q used = 8.33%)



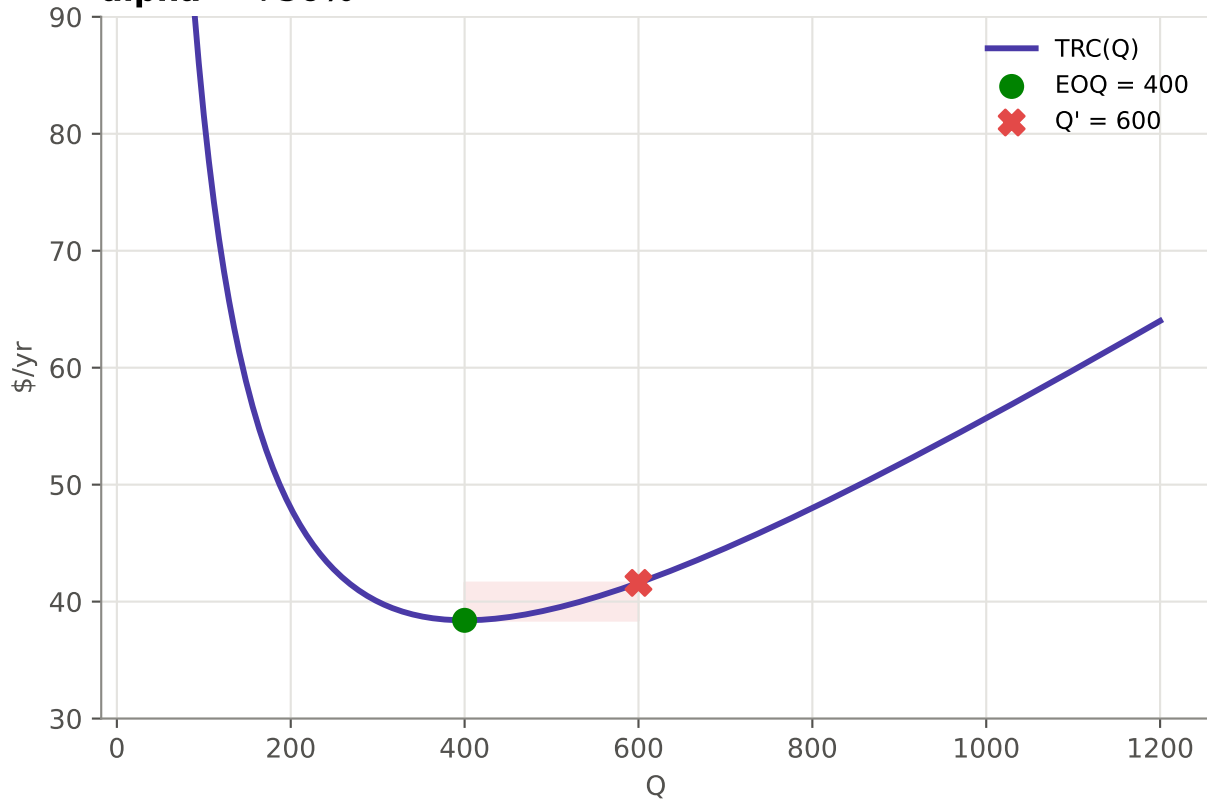
## EOQ sawtooth



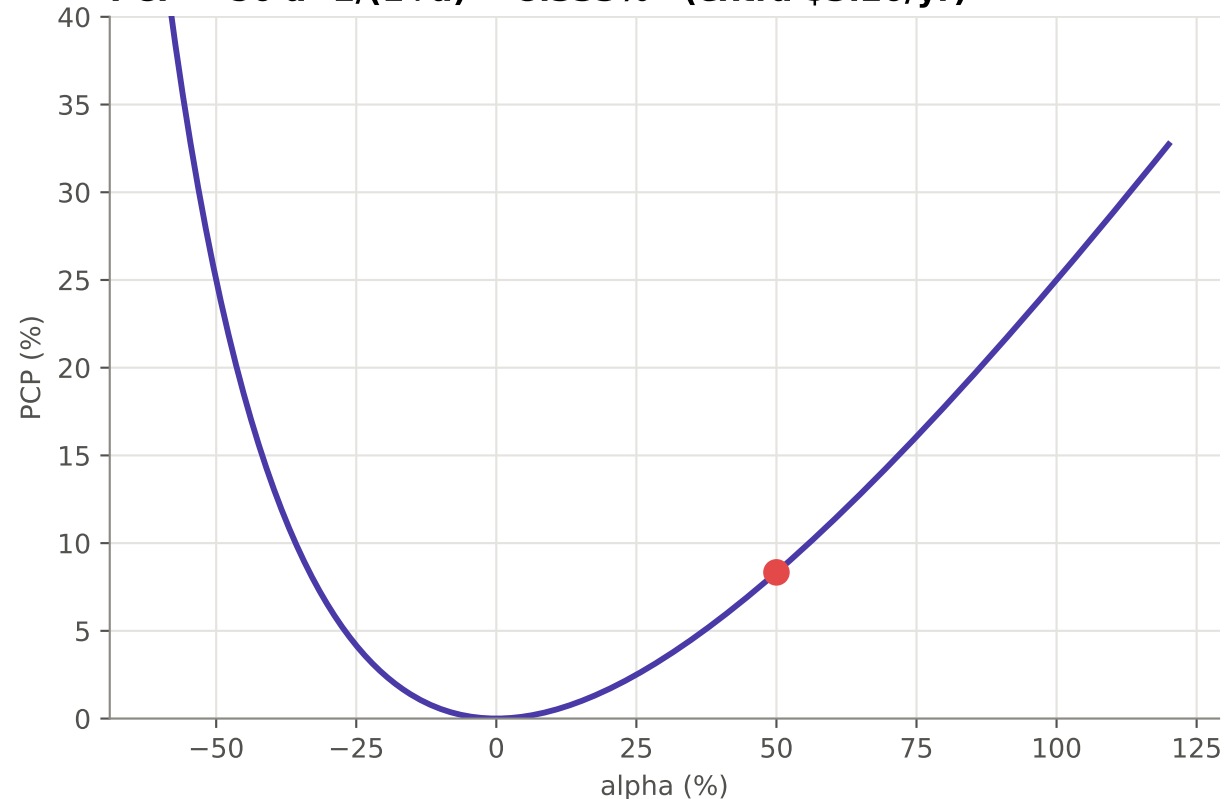
## Sensitivity: practical policies

| scenario                               | Q_used | alpha  | PCP (%) | Extra cost (\$/yr) |
|----------------------------------------|--------|--------|---------|--------------------|
| Order exactly the EOQ                  | 400    | 0      | 0       | 0                  |
| Supplier ships in cases of 250         | 250    | -0.375 | 11.25   | 4.32               |
| Supplier ships in cases of 500         | 500    | 0.25   | 2.5     | 0.96               |
| Buyer orders one month of demand       | 200    | -0.5   | 25      | 9.6                |
| Buyer orders one quarter of demand     | 600    | 0.5    | 8.333   | 3.2                |
| Warehouse orders half a year at a time | 1,200  | 2      | 66.67   | 25.6               |
| Buyer orders weekly                    | 46     | -0.885 | 340.5   | 130.8              |

**alpha = +50%**



**PCP = 50  $a^2/(1+a)$  = 8.333% (extra \$3.20/yr)**

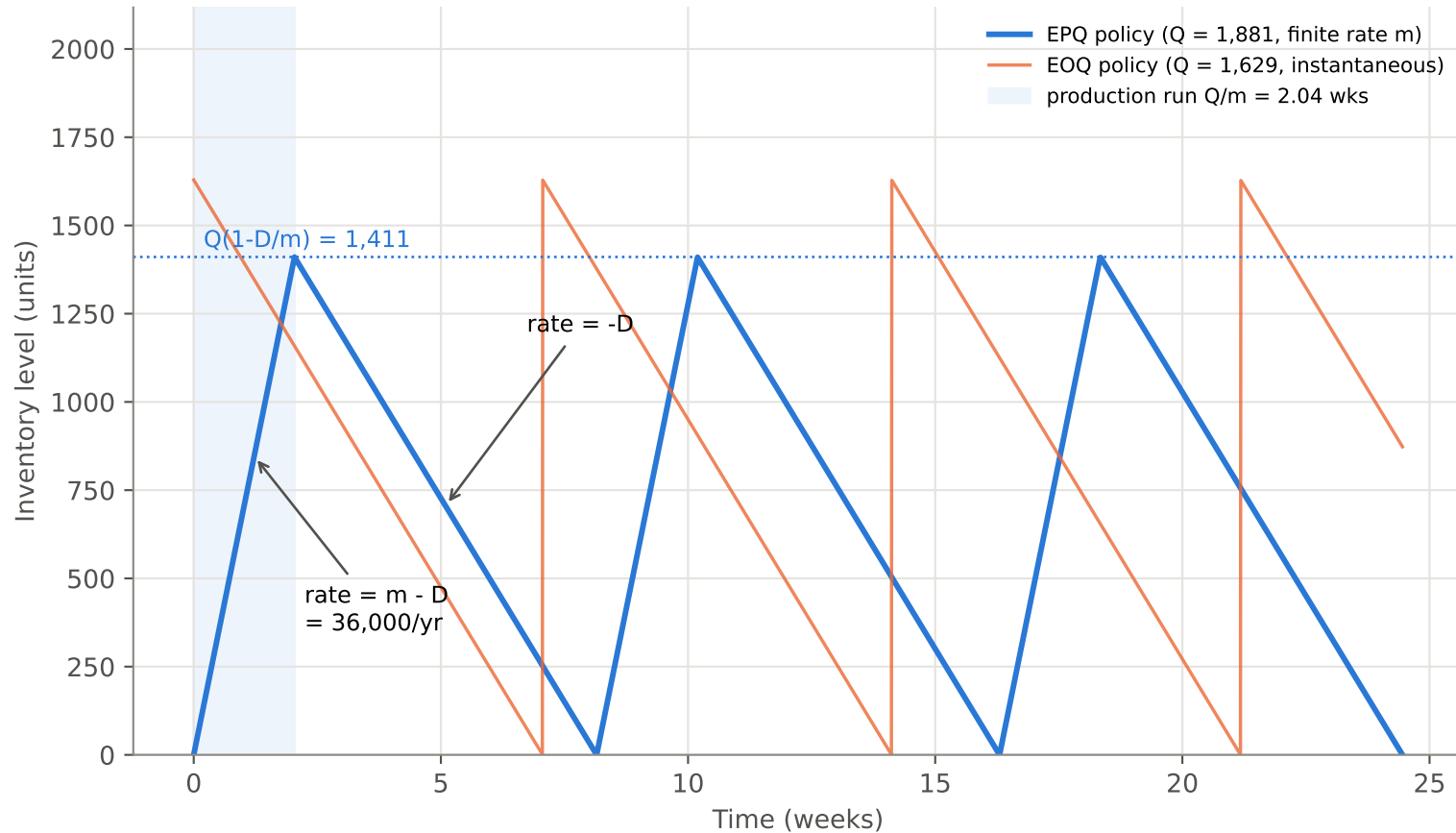


## Worked example 2: EPQ

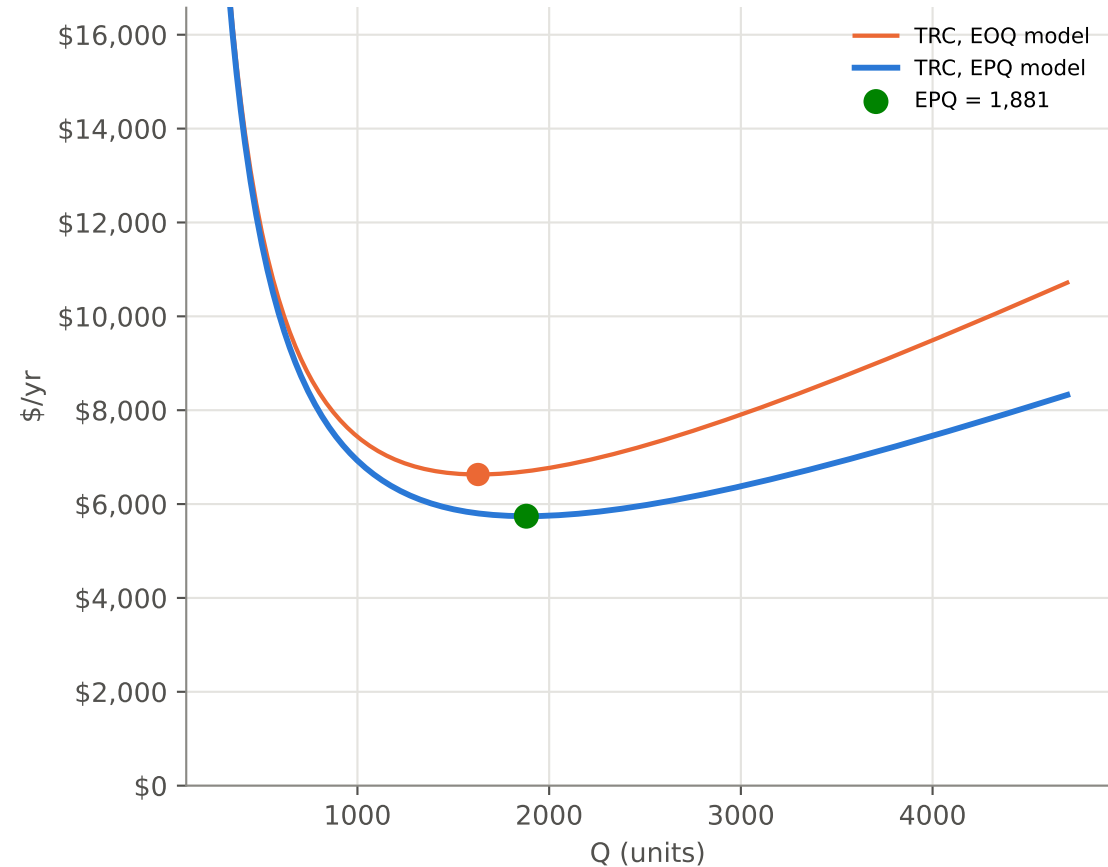
| Quantity                             | Value |
|--------------------------------------|-------|
| EPQ (units)                          | 1,881 |
| EOQ for comparison (units)           | 1,629 |
| $EPQ / EOQ = 1/\sqrt{1-D/m}$         | 1.155 |
| TRC(EPQ) (\$/yr)                     | 5,742 |
| Setup cost $AD/Q$ (\$/yr)            | 2,871 |
| Carrying cost (\$/yr)                | 2,871 |
| Maximum inventory $Q(1-D/m)$ (units) | 1,411 |
| Average inventory (units)            | 705.4 |
| Production runs per year $D/Q$       | 6.38  |
| Cycle length $Q/D$ (weeks)           | 8.151 |
| Production run length $Q/m$ (weeks)  | 2.038 |
| Line idle time per cycle (weeks)     | 6.113 |
| Utilisation of the line $D/m$        | 0.25  |



**Inventory profile: finite production rate vs instantaneous delivery**



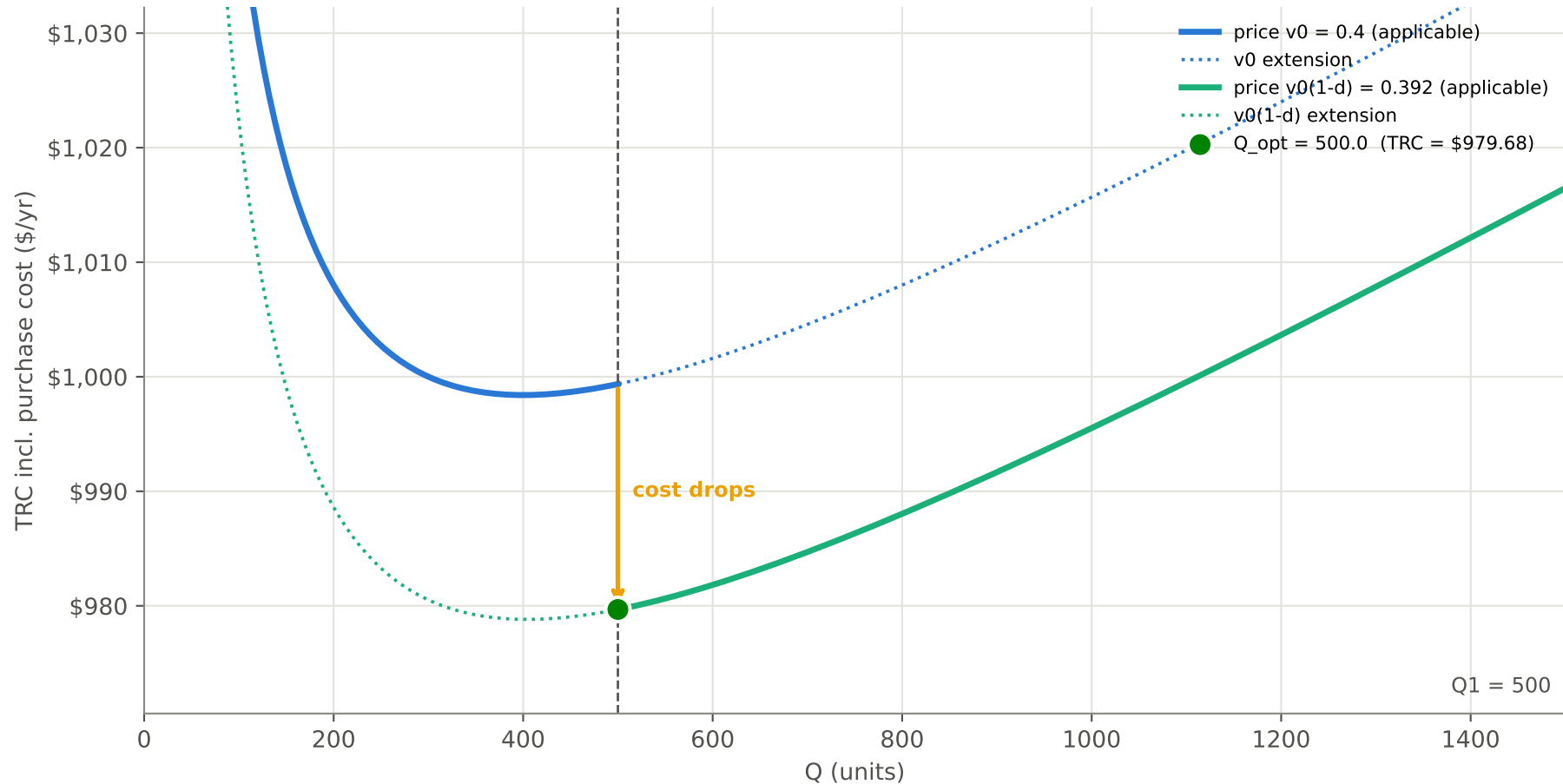
**EPQ/EOQ =  $1/\sqrt{1-D/m} = 1.155$**



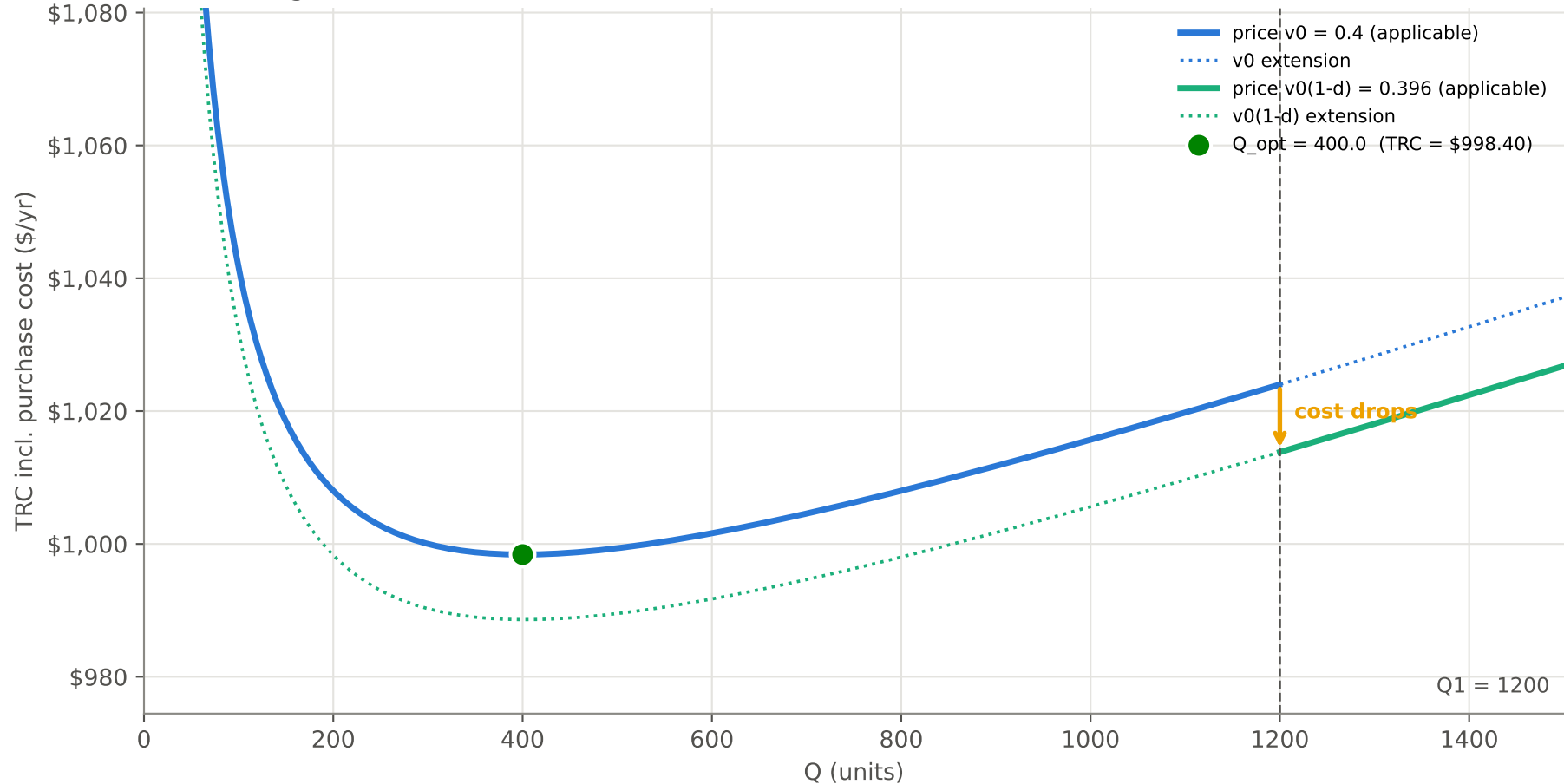
## Worked example 3: quantity discount cases

| Q1    | d    | EOQ_d | EOQ_0 | TRC_EOQ0 | TRC_Q1 | case | Q_opt | TRC_opt | saving_vs_no_disc |
|-------|------|-------|-------|----------|--------|------|-------|---------|-------------------|
| 500   | 0.02 | 404.1 | 400   | 998.4    | 979.7  | 1    | 500   | 979.7   | 18.72             |
| 1,200 | 0.01 | 402   | 400   | 998.4    | 1,014  | 2    | 400   | 998.4   | 0                 |
| 300   | 0.02 | 404.1 | 400   | 998.4    | 980.5  | 3    | 404.1 | 978.8   | 19.59             |

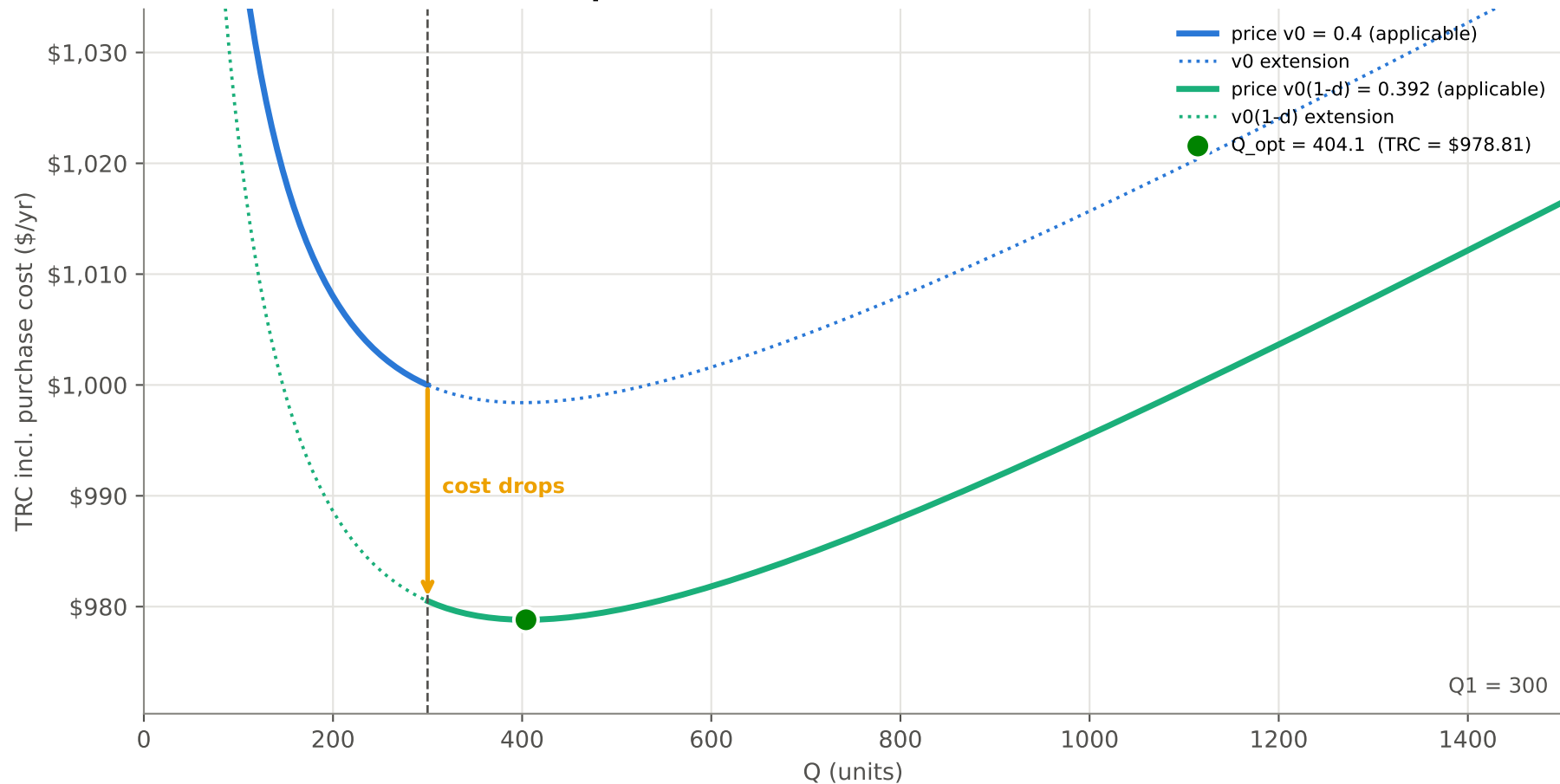
### Case 1 - break quantity is optimal



## Case 2 - ignore the discount



### Case 3 - discounted EOQ is optimal



**Discount decision map ( $A=3.2$ ,  $D=2,400$ ,  $v_0=0.4$ ,  $r=0.24$ )**

